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AAJ Global Future and Emerging Tech Guidebook

GUIDEBOOK

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Indented Use

The intended use of the AAJ Global Future and Emerging Tech index is to provide investors with exposure to innovative companies at the forefront of global technological advancement, particularly those with substantial capital dedicated to research and development (R&D). By focusing on large-cap companies engaged in transformative sectors such as 3D printing, AI ecosystems, blockchain, and green energy utilities, the index seeks to capture opportunities created by emerging technologies poised for substantial growth. Additionally, the index aims to offer exposure to companies demonstrating high returns on invested capital (ROIC), indicating their efficiency in converting capital into profits and their potential for sustainable growth. Through its selection criteria and equal-weighting methodology, the index aims to provide a balanced approach to investing in innovative companies, mitigating concentration risk while maximizing exposure to small up-and-coming firms with significant growth potential. Overall, the index serves as a vehicle for investors looking to capitalize on the inevitable upward trajectory of new technology across diverse industries, thereby enhancing their investment portfolios with opportunities for long-term growth and innovation.

General Information

Name:	AAJ Global Future and Emerging Tech
Currency:	The index is denominated in USD (United States Dollar).
Return Calculation:	The return calculation is based on the total return, which includes both the capital appreciation/depreciation and any distributions such as dividends.
Number of Constituents:	62
Base Universe:	Global Companies in frontier tech industries

Base Methodology:

Eligible Universe: Global companies

Filter for frontier Companies in: 3d Printing, 5g, AI Ecosystem, Big Data, Blockchain, Cloud Computing, Genome Editing, Nanotechnology, Laser Technology, Robotics, Virtual and Augmented Reality, Agricultural Biotech, Cyber Security, Electric Vehicles, Green Energy Utilities, Human Augmentation Technology, Medical Biotech, Metaverse, NFSs, Social Media, Clean Tech, Energy Tech

Market Cap Filter: Filter out companies with market cap less than 33,000 Millio USD

Quality Filter: Filter out companies with return on invested capital less than 20%

Weighting: Equally weight the items in the index. Under alternative methods the index experiences severe overweighting in a few companies, leading to up to 50% weight in 5 companies.

Caps / Floors: No caps or floors are implemented; the equal weighting prevents overinvestment in a single security

Reconstitution Schedule: The index selects constituents Quarterly in March-June-September-December. The ex-date is the 3rd Friday of each Reconstitution Month. The Determination Date for Ordinary Adjustments is 10 days before the Ex-date

Rebalancing Schedule: The index is rebalanced on a quarterly basis in March-June-September-December. The ex-date is the 3rd Friday of each Rebalancing Month. The Determination Date for Ordinary Adjustments is 10 days before the Ex-date

Exit Buffer: There is a 5% buffer to keep constituents in the index. This helps to reduce constant add and removing of a company end the end of its rapid growth period

Fast Entry / Exit: There is no fast entry/exit implementation

Index Calculation

The index value is calculated using the following formula:

$$Index_t = \sum_{i}^{n} \frac{P_{i,t} * IWF_{i,t} * TOS_{i,t} * AWF_{i,t} * F_{i,t}}{D_t}$$

Pi,t= Price of the constituent ‘i’ at time ’t’

IWFi,t=Investable Weighting Factor of constituent ‘i’ at time ’t’

TOSi,t= Total no. of Shares Outstanding of constituent ‘i’ at time ’t’

AWFi,t= Adjustable Weighting Factor for Equal weighted constituent ‘i’ at time ’t’

Fi,t= Foreign exchange Rate for constituent ‘i’ at time ’t’

Dt=Divisor Value at time ’t’

On each Adjustment Day t, the divisor is adjusted to keep the index value constant. The new divisor is calculated as:

$$D_{t+1} = D_t + \frac{Change\ in\ Market\ Cap\ of\ the\ Index}{Index\ Value_t}$$

The new divisor is then used in the calculation of the following day's index open.

Index Benchmark:

The benchmark for the analysis is the iShares Edge MSCI USA Quality Factor ETF. This ETF is used as a standard to compare the performance and risk attributes of the AAJ Global Future and Emerging Tech index.

Index Characteristics

As of 2024-04-18, the index had the following key characteristics:

- 62 constituents
- Median constituent weight: 1.61%
- Maximum constituent weight: 2.08%
- Minimum constituent weight: 1.32%
- Total market capitalization: $20.1 trillion
- Mean market capitalization: $324.2 billion

Performance Attribution:

	Relative Allocation	Return in Portfolio	Allocation Effect	Selection Effects
Energy	-0.0364	-0.0002	-0.0436	-0.0502
Materials	0.0814	0.0996	0.054	0.0068
Utilities	-0.0294	0	-0.0079	-0.0079
Financials	-0.1296	0.0443	-0.1254	0.0288
Health Care	0.169	0.3064	0.1404	0.0217
Industrials	-0.0576	0.0143	-0.0493	-0.0438
Real Estate	-0.0289	0	-0.0156	-0.0156
Consumer Staples	-0.0741	0	-0.0378	-0.0378
Communication Services	-0.0391	0.0857	-0.048	0.0169
Consumer Discretionary	-0.0784	0.0193	-0.075	-0.0009
Information Technology	0.2231	0.6001	0.2824	0.0307

Relative Allocation Return: This refers to the return generated by deviating from the benchmark weights for each sector in the portfolio. A negative value suggests that the portfolio's allocation to the sector underperformed compared to the benchmark, while a positive value indicates outperformance.

Portfolio Allocation Effect: This is the cumulative effect of the relative allocation returns across all sectors. It represents the overall impact of the portfolio manager's decisions to deviate from the benchmark weights.

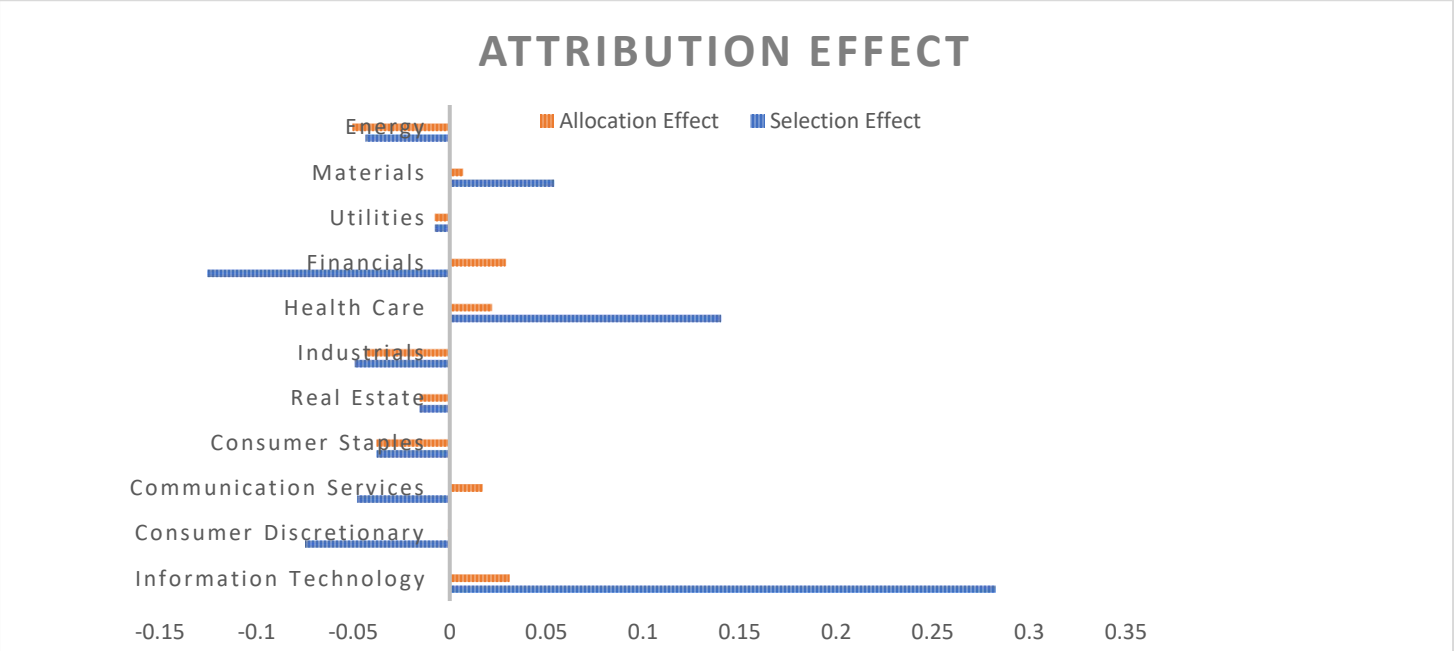
Selection Effect: This component reflects the return generated by the specific securities chosen within each sector compared to the benchmark. It measures the performance difference between the individual securities in the portfolio and their counterparts in the benchmark.

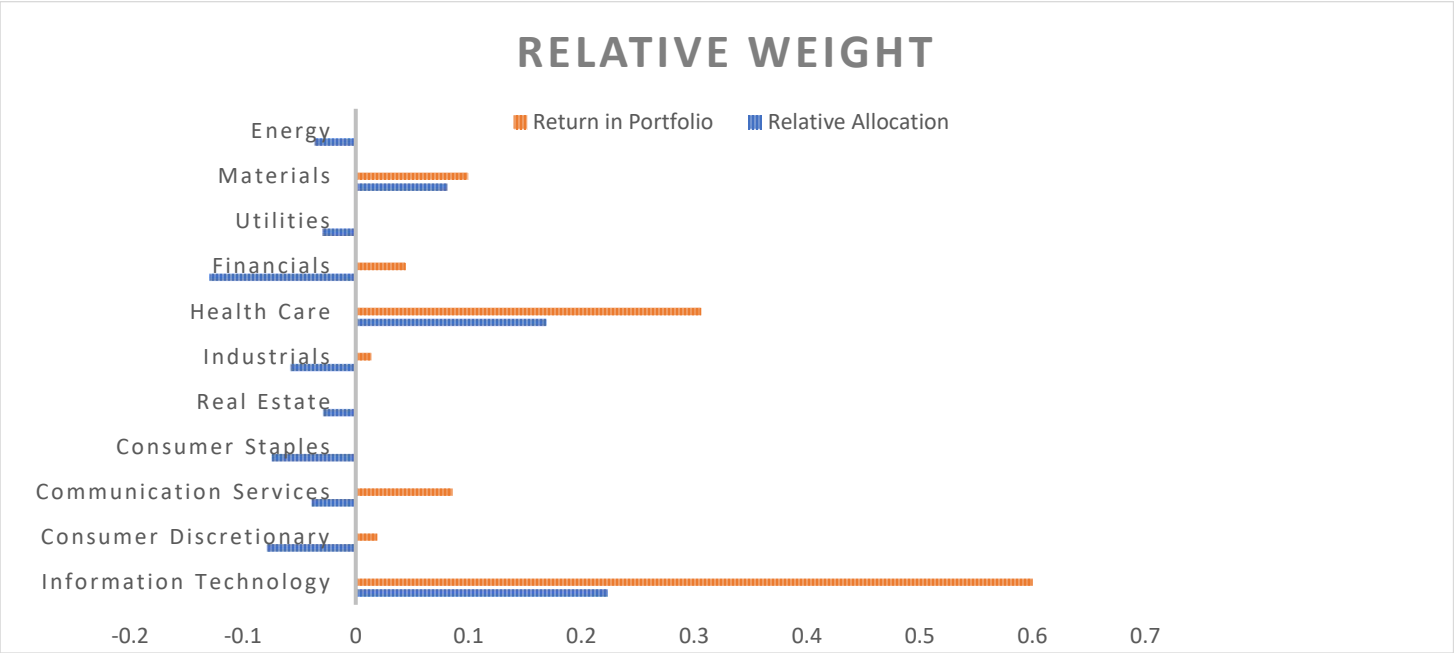
Return in Portfolio: This is the actual return achieved within the portfolio for each sector. It includes both the relative allocation return and the selection effect.

Performance Attribution by GICS Sectors (5 Years)

Over a five-year period, the performance attribution analysis provides important insights into the ways in which various Global Industry Classification Standard (GICS) sectors influenced the index's performance. Especially:

- **Information Technology** with a relative allocation effect of 0.2824 and a selection effect of 0.0307, demonstrated a remarkably favourable impact, demonstrating great sector performance and wise stock selection.
- **Health Care** also contributed positively with a relative allocation effect of 0.1404 and a selection effect of 0.0217.
- **Financials** and **Energy** sectors had negative allocation effects, indicating underperformance or underweight positions relative to the benchmark.





Energy: The portfolio underperformed compared to the benchmark in terms of allocation (-0.0364) and selection (-0.0502), resulting in a negative overall portfolio allocation effect (-0.0436).

Materials: The portfolio outperformed both in terms of allocation (0.0814) and selection (0.0068), resulting in a positive overall portfolio allocation effect (0.054).

Utilities: There was a slight underperformance compared to the benchmark in terms of allocation (-0.0294), but no selection effect, resulting in a negative overall portfolio allocation effect (-0.0079).

Financials: The portfolio significantly underperformed in terms of allocation (-0.1296), but outperformed in terms of selection (0.0288), resulting in a negative overall portfolio allocation effect (-0.1254).

Health Care: The portfolio significantly outperformed in both allocation (0.169) and selection (0.0217), resulting in a positive overall portfolio allocation effect (0.1404).

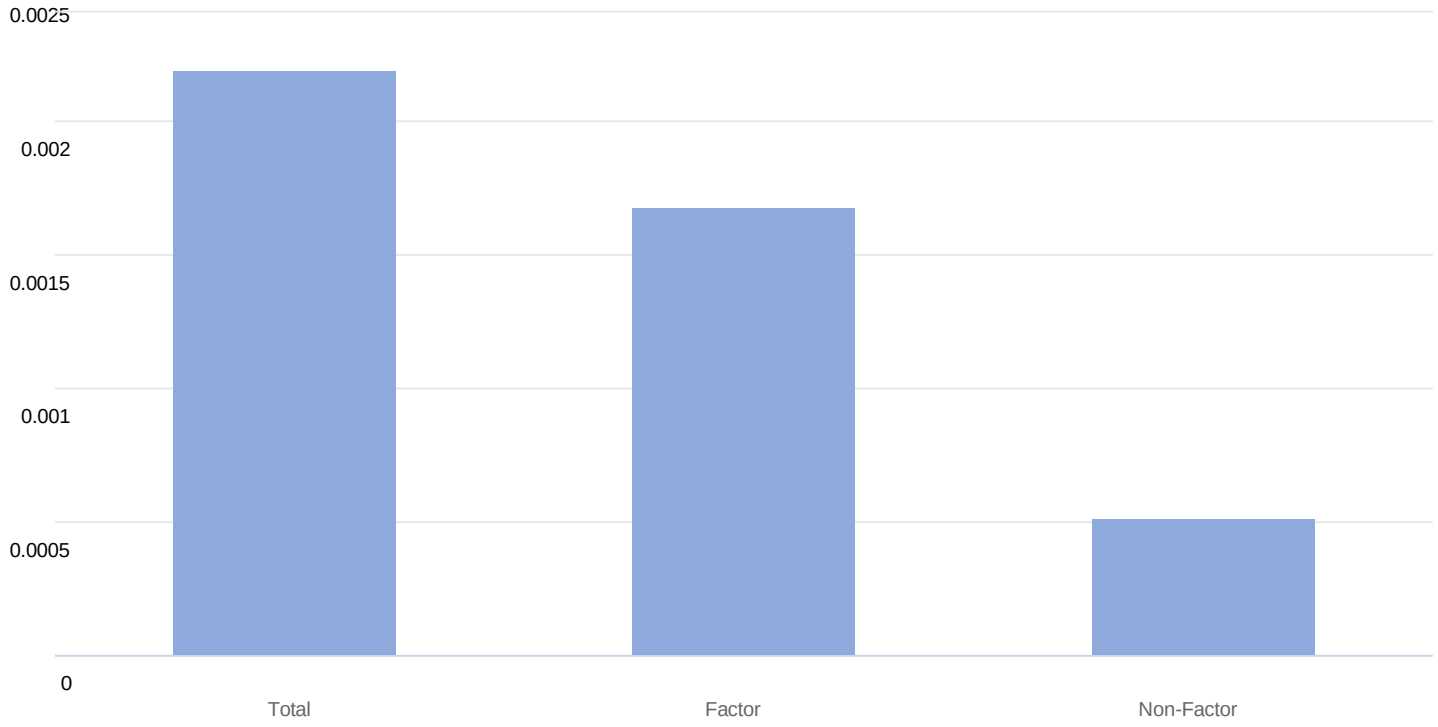
Industrials: The portfolio underperformed in terms of allocation (-0.0576) and selection (-0.0438), resulting in a negative overall portfolio allocation effect (-0.0493).

Real Estate, Consumer Staples, Communication Services, and Consumer Discretionary: These sectors all experienced underperformance in terms of allocation and/or selection, leading to negative overall portfolio allocation effects.

Information Technology: This sector saw significant outperformance in both allocation (0.2231) and selection (0.0307), resulting in a positive overall portfolio allocation effect (0.2824).

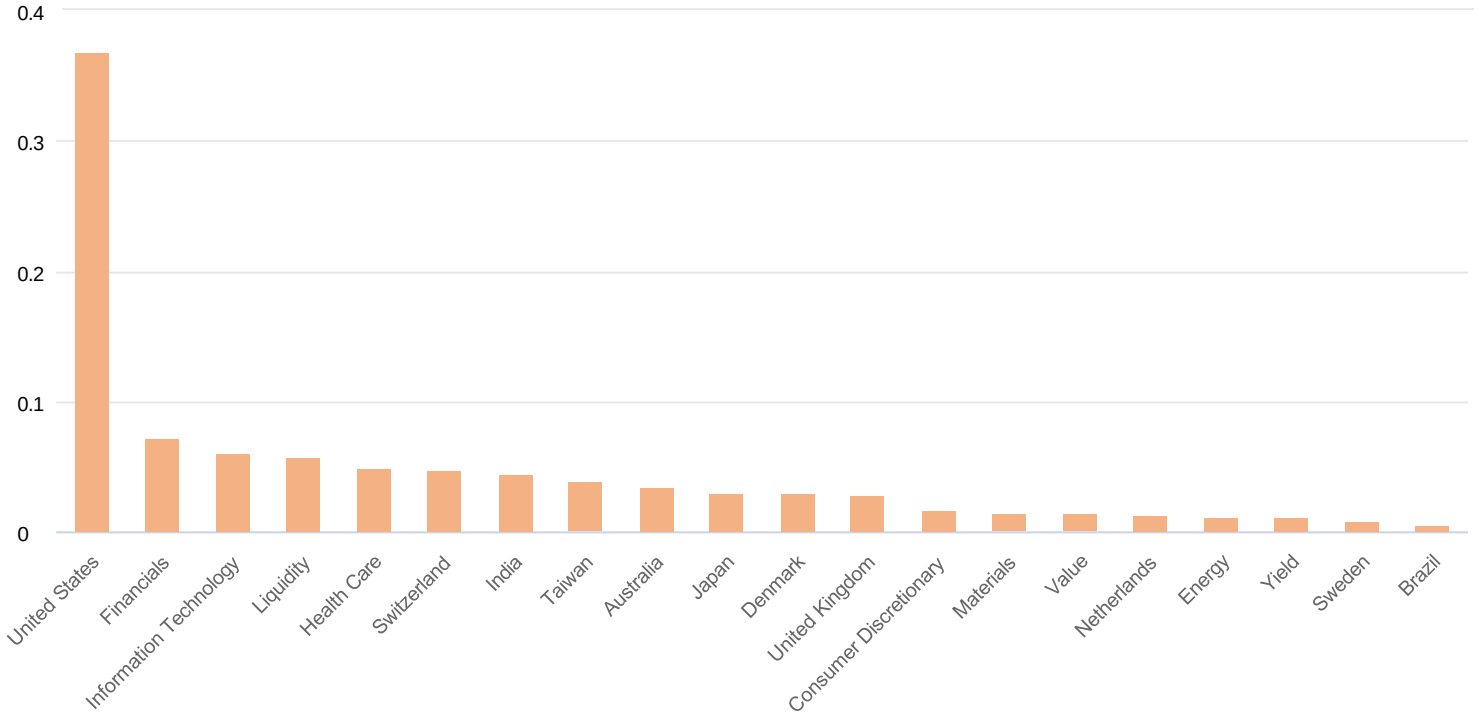
Risk Factor Attribution (5 Years)

Active Risk



Active Factor Risk Breakdown

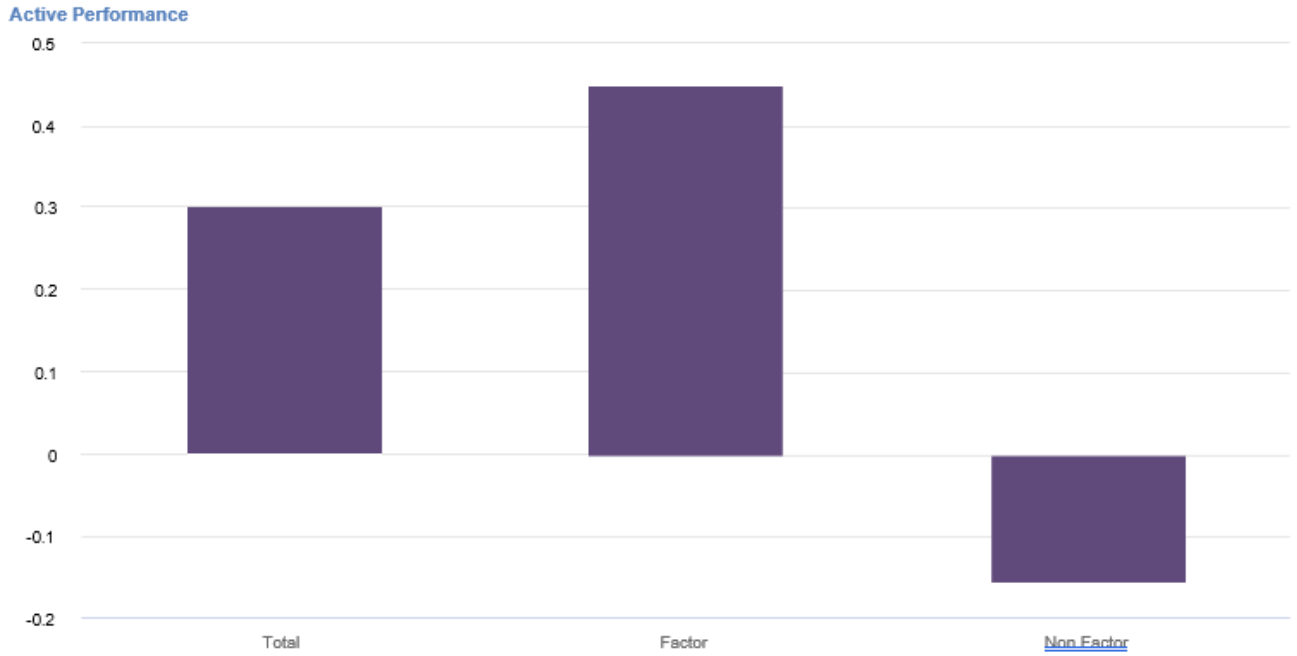
The active factor risk breakdown further delves into the specifics of risk contributions by different factors, providing a nuanced understanding of the index's risk profile.



- Geographical factors such as the United States, Switzerland, and India showed varying degrees of risk contribution.
- Sectoral factors like Financials and Information Technology also contributed to the active risk profile.
- Style factors such as Value and Yield were identified, indicating their influence on the risk dynamics of the index.

Performance Factor Attribution

The performance factor attribution section analyses the active performance and its breakdown:



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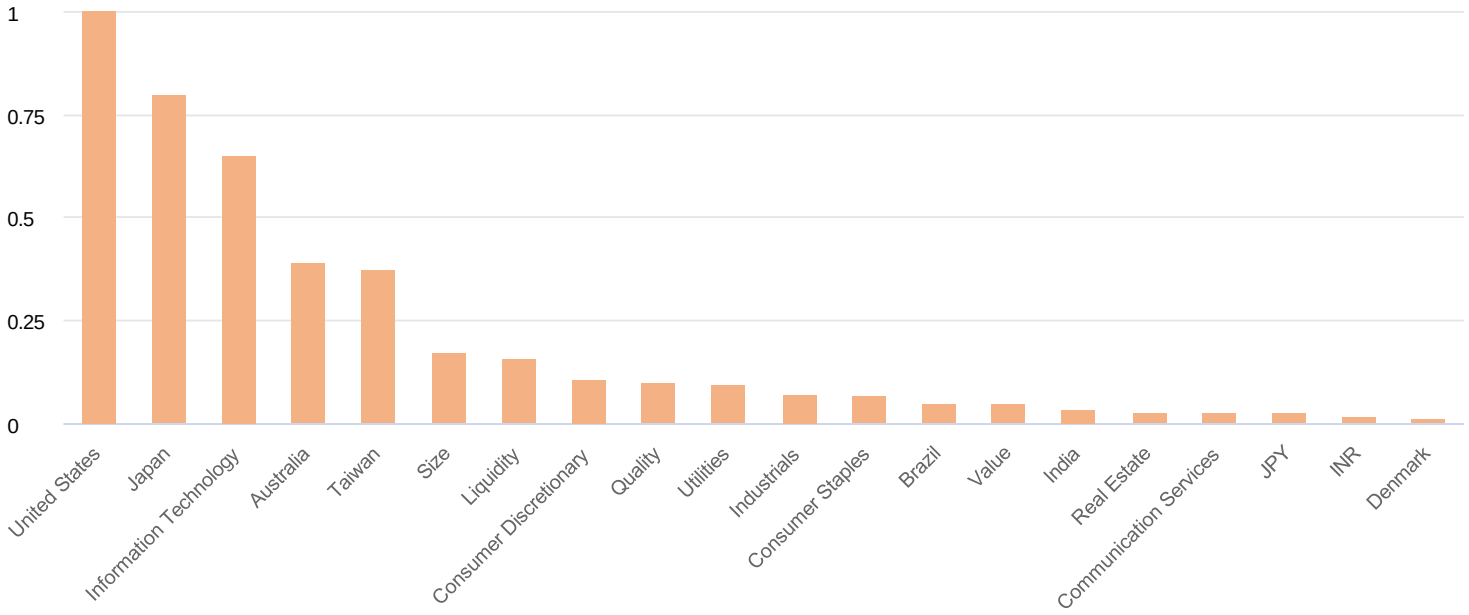
Active Factor Performance Breakdown



The performance factor attribution section analyses the active performance and its breakdown:

- Geographical contributions from the United States and Japan were notable.
- Sectoral contributions from Information Technology and Consumer Discretionary sectors were significant.
- Style factors like Size, Liquidity, and Quality also played crucial roles in the index's performance.

Style Factor Tilts

The style factor tilts as of the end date (2024-04-18) compare the index against the benchmark across various factors:

	Index	Benchmark
Momentum	-0.11	0.06
Value	-0.06	0.10
Size	0.53	0.51
Quality	0.30	0.42
Low Volatility	-0.40	0.02
Liquidity	0.59	0.63
Yield	-0.22	-0.27

- **Momentum**, **Value**, and **Low Volatility factors** showed divergences between the index and the benchmark, indicating strategic tilts.
- **Size** and **Liquidity factors** were more aligned, suggesting similar exposures to these style factors between the index and the benchmark.

Conclusion

In summary A thorough examination of the AAJ Global Future and Emerging Tech index, benchmarked against the iShares Edge MSCI USA Quality Factor ETF, is provided in the Risk and Factor Attribution Report. The research provides insights into the index's strategic positioning, performance drivers, and risk profile over a 5-year period through thorough performance attribution by GICS sectors, risk factor attribution, and style factor tilts. The dynamics of the index can be better understood and future investment choices can be made with the help of this study.